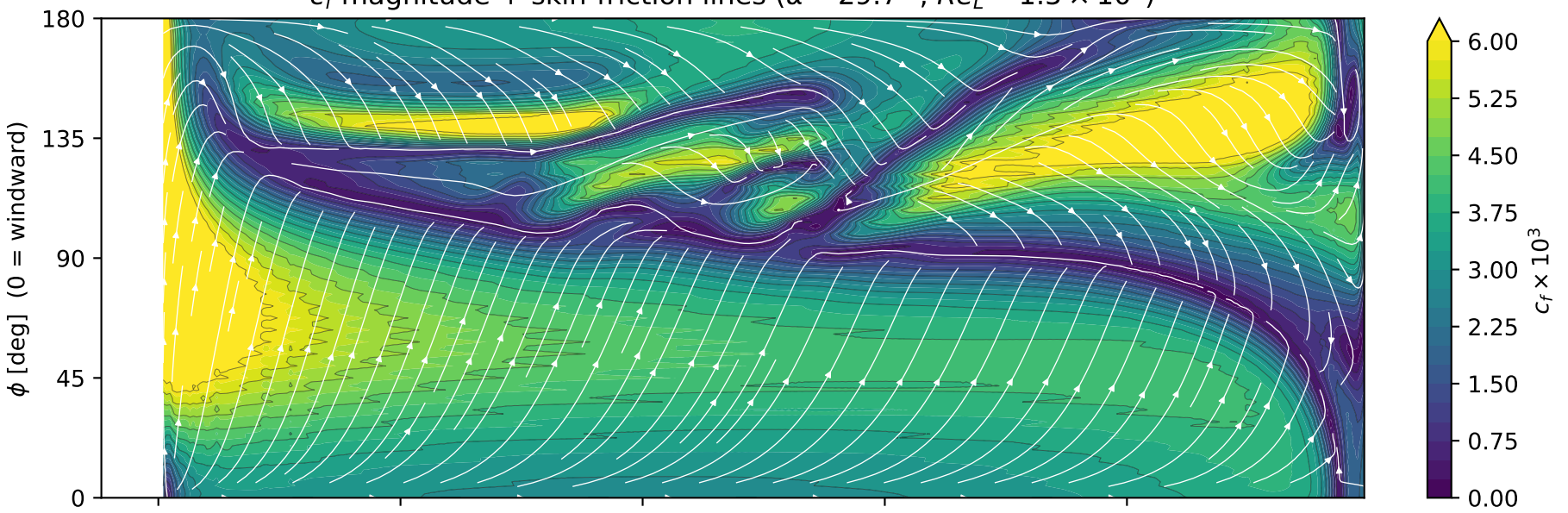
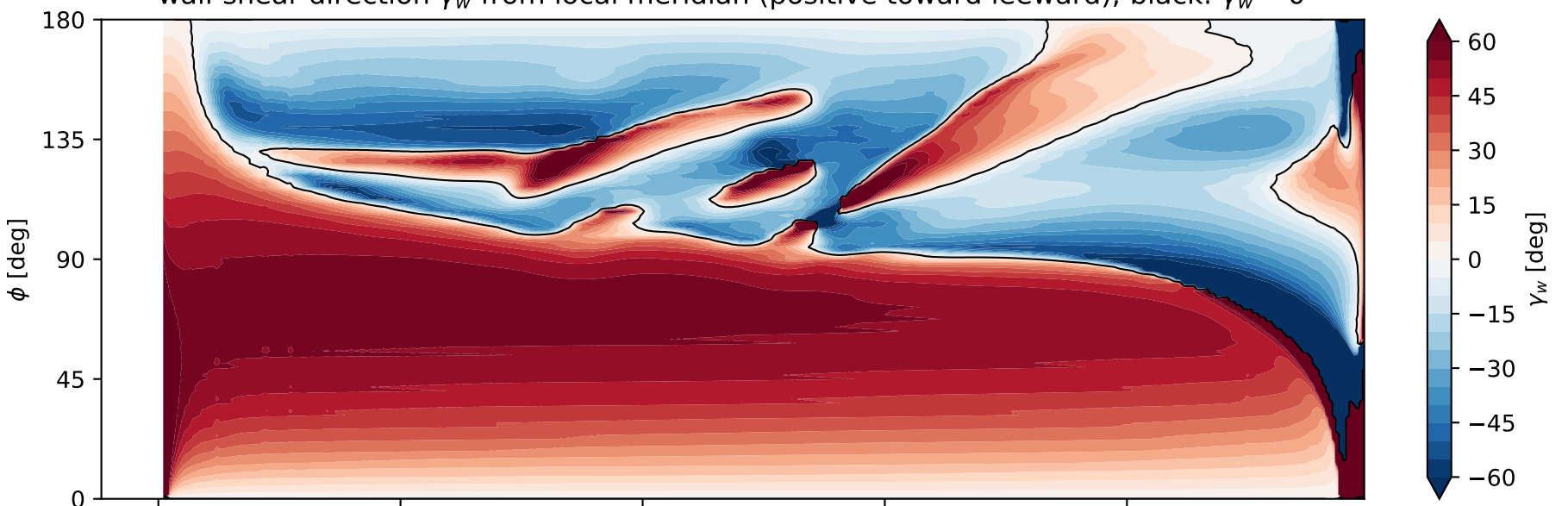


c_f magnitude + skin-friction lines ($\alpha = 29.7^\circ$, $Re_L = 1.5 \times 10^6$)



wall-shear direction γ_w from local meridian (positive toward leeward); black: $\gamma_w = 0$



near-wall $\max \chi$; cyan: $\chi = c_{v1}$ (model-native transition front)

